MAT 101 - Linear Algebra & Calculus (2019 scheme)

Answer Key

Part A

(1) Find the rank of $A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 2 & 2 & 4 & 1 \\ 5 & 6 & 7 & 5 \end{bmatrix}$

Sol: $R_2 : R_2 - 2R_1$, $R_3 : R_3 - 5R_1$ $\Rightarrow A \sim \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 6 & -5 \\ 0 & -4 & 12 & -10 \end{bmatrix}$

$R_3 : R_3 - 2R_2 \Rightarrow A = \begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & -2 & 6 & -5 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$\Rightarrow$ $\underline{\text{Rank } A = 2}$ (no. of non-zero rows)

(2) What type of conic section the quadratic form represents?

$Q = 17x_1^2 - 30x_1x_2 + 17x_2^2 = 128$

Sol: Given symmetric matrix is $A = \begin{bmatrix} 17 & -15 \\ -15 & 17 \end{bmatrix}$

$|A - \lambda I| = 0 \Rightarrow \begin{vmatrix} 17-\lambda & -15 \\ -15 & 17-\lambda \end{vmatrix} = 0 \Rightarrow \lambda = 2, 32$

Canonical form is: $2y_1^2 + 32y_2^2 = 128 \Rightarrow$ Ellipse

(3) If $u = \dfrac{x^3+y^3}{x-y}$ find $\dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y}$.

Sol: $\dfrac{\partial u}{\partial x} = \dfrac{(x-y)\cdot 3x^2 - (x^3+y^3)}{(x-y)^2} = \dfrac{2x^3 - 3x^2y - y^3}{(x-y)^2}$

$\dfrac{\partial u}{\partial y} = \dfrac{(x-y)\,3y^2 - (x^3+y^3)\times -1}{(x-y)^2} = \dfrac{x^3 + 3xy^2 - 2y^3}{(x-y)^2}$

$\therefore \dfrac{\partial u}{\partial x} + \dfrac{\partial u}{\partial y} = \dfrac{3x^3 - 3y^3 - 3x^2y + 3xy^2}{(x-y)^2}$

(4) If $z = x^2 y$, $x = t^2$, $y = t^3$, find $\frac{dz}{dt}$ using chain rule.

Sol.: $z \to (x,y) \to t$

$$\frac{dz}{dt} = \frac{\partial z}{\partial x}\cdot\frac{dx}{dt} + \frac{\partial z}{\partial y}\cdot\frac{dy}{dt} = 2xy\cdot 2t + x^2\cdot 3t^2$$
$$= 2t^2 t^3 \cdot 2t + (t^2)^2\cdot 3t^2 = 4t^6 + 3t^6 = \underline{\underline{7t^6}}$$

(5) Evaluate $\int_{z=0}^{3}\int_{y=0}^{2}\int_{x=0}^{1} xyz\,dx\,dy\,dz$

Sol.: $$\int_{z=0}^{3} z\,dz \times \int_{y=0}^{2} y\,dy \times \int_{x=0}^{1} x\,dx = \left(\frac{z^2}{2}\right)_{z=0}^{3}\cdot\left(\frac{y^2}{2}\right)_{y=0}^{2}\cdot\left(\frac{x^2}{2}\right)_{x=0}^{1}$$
$$= \frac{9}{2}\cdot\frac{4}{2}\cdot\frac{1}{2} = \underline{\underline{\frac{9}{2}}}$$

(6) Use double integral to find the volume of the solid enclosed below the plane $z = 4-x-y$ and above the rectangle $R = \{(x,y);\ 0\le x\le 1,\ 0\le y\le 2\}$

Sol.: $$V = \iint z\,dA = \int_{y=0}^{2}\int_{x=0}^{1}(4-x-y)\,dx\,dy = \int_{y=0}^{2}\left(4x-\frac{x^2}{2}-yx\right)_{x=0}^{1} dy = \cdots = \underline{\underline{5}}$$

(7) Does the series $\sum_{k=1}^{\infty}\left(\frac{4}{5}\right)^k$ converge? If so find the sum.

Sol.: $\left(\frac{4}{5}\right)^1 + \left(\frac{4}{5}\right)^2 + \left(\frac{4}{5}\right)^3 + \cdots$; $r = \frac{4}{5} < 1 \Rightarrow$ Converges. $S = \frac{a}{1-r}$

$$\therefore \text{Sum} = S = \frac{4/5}{1-4/5} = \frac{4/5}{1/5} = \frac{4}{5}\times\frac{5}{1} = \underline{\underline{4}}$$

(8) Test the convergence of the series $\sum_{k=1}^{\infty}\frac{k^2}{2k^2-1}$

Sol.: $$\lim_{k\to\infty} a_k = \lim_{k\to\infty}\frac{k^2}{2k^2-1} = \frac{k^2}{k^2(2-1/k^2)} = \frac{1}{2} \neq 0 \Rightarrow \text{diverges.}$$

(9) Find the binomial series for $f(x) = \frac{1}{\sqrt{1+x}}$ upto 3rd degree terms;

Sol.: $$(1+x)^n = 1 + \frac{nx}{1!} + \frac{n(n-1)}{2!}x^2 + \cdots$$
$$(1+x)^{-1/2} = 1 - \frac{1}{2}x + \frac{1.3}{2^2 2!}x^2 + \frac{1.3.5}{2^3.3!}x^3 + \cdots$$

(10) Find the Maclaurin's series for $f(x) = x\cos x$ upto 3rd degree term

Sol: $f(x) = f(0) + \frac{x}{1!} f'(0) + \frac{x^2}{2!} f''(0) + \cdots$

$f(x) = x\cos x \quad , \quad f(0) = 0$

$f'(x) = x\cdot\bar{\sin x} + \cos x \quad , \quad f'(0) = 1$

$f''(x) = \bar{}[x\cdot\cos x + \sin x] \bar{+} \sin x \quad , \quad f''(0) = 0$

$f'''(x) = \bar{}[x\cdot\bar{\sin x} + \cos x + \cos x] - \cos x \quad , \quad f'''(0) = \bar{3}$

$\therefore \; x\cos x = \frac{x}{1!} + \frac{x^3}{2!} + - \cdots$

Part B

11 (a) Using Gauss elimination method, find the solution of the system $x+y-z=9$, $8y+6z=-6$, $-2x+4y-6z=40$

Sol: $[A:B] = \begin{bmatrix} 1 & 1 & -1 & 9 \\ 0 & 8 & 6 & -6 \\ -2 & 4 & -6 & 40 \end{bmatrix}$

$R_3 : R_3 + 2R_1 \Rightarrow [A:B] = \begin{bmatrix} 1 & 1 & -1 & 9 \\ 0 & 8 & 6 & -6 \\ 0 & 6 & -8 & 58 \end{bmatrix}$

$R_3 : 4R_3 - 3R_2 \Rightarrow [A:B] = \begin{bmatrix} 1 & 1 & -1 & 9 \\ 0 & 8 & 6 & -6 \\ 0 & 0 & -50 & 250 \end{bmatrix}$

$\Rightarrow \left.\begin{aligned} x+y-z &= 9 \\ 8y+6z &= -6 \\ -50z &= 250 \end{aligned}\right\} \Rightarrow \underline{\underline{z = -5, \; y = 3, \; x = 1}}$

11 (b) Find the matrix of transformation that diagonalize the matrix $\begin{bmatrix} 3 & 1 & -1 \\ -2 & 1 & 2 \\ 0 & 1 & 2 \end{bmatrix}$. Also find the diagonal matrix.

Sol: characteristic equ: $|A - \lambda I| = 0 \Rightarrow \lambda^3 - 6\lambda^2 + 11\lambda - 6 = 0$

solving : $\lambda = 1, 2, 3$.

Eigen vectors are $\begin{bmatrix}1\\-1\\1\end{bmatrix}$, $\begin{bmatrix}1\\0\\1\end{bmatrix}$, $\begin{bmatrix}0\\1\\1\end{bmatrix}$. Thus, matrix of transformation is

$$B = \begin{bmatrix}1 & 1 & 0\\-1 & 0 & 1\\1 & 1 & 1\end{bmatrix} \text{ and } D = \begin{bmatrix}1 & 0 & 0\\0 & 2 & 0\\0 & 0 & 3\end{bmatrix}$$

---

12 (a) Find the value of λ & μ for which the system of equations $2x+3y+5z=9$, $7x+3y-2z=8$, $2x+3y+\lambda z=\mu$ has (i) no solution (ii) unique solution (iii) more than one solution.

Sol: $[A:B] = \begin{pmatrix}2 & 3 & 5 & 9\\7 & 3 & -2 & 8\\2 & 3 & \lambda & \mu\end{pmatrix}$, Using suitable row operations,

$$[A:B] \sim \begin{bmatrix}2 & 3 & 5 & 9\\0 & -15 & -39 & -47\\0 & 0 & \lambda-5 & \mu-9\end{bmatrix}$$

(i) no soln. $\lambda=5$, $\mu\neq 9$

(ii) unique soln: $\lambda\neq 5$, $\mu=$ any value

(iii) more soln: $\lambda=5$, $\mu=9$

---

12 (b) Find the eigen values and eigen vectors of $\begin{bmatrix}-2 & 2 & -3\\2 & 1 & -6\\-1 & -2 & 0\end{bmatrix}$

Sol: $|A-\lambda I|=0 \Rightarrow \lambda^3+\lambda^2-21\lambda-45=0 \Rightarrow \lambda = 5, -3, -3$

$\lambda=5 \Rightarrow X_1 = \begin{bmatrix}1\\2\\-1\end{bmatrix}$

$\lambda=-3 \Rightarrow x-2y+3z=0 \Rightarrow X_2 = \begin{bmatrix}2\\1\\0\end{bmatrix}$, $X_3 = \begin{bmatrix}-3\\0\\1\end{bmatrix}$

---

13 (a) If $w = \sqrt{x^2+y^2+z^2}$, $x=\cos\theta$, $y=\sin\theta$, $z=\tan\theta$ find $\frac{dw}{d\theta}$ at $\theta=\pi/4$

$w \rightarrow (x,y,z) \rightarrow \theta$

Sol:
$$\frac{dw}{d\theta} = \frac{\partial w}{\partial x}\cdot\frac{dx}{d\theta} + \frac{\partial w}{\partial y}\cdot\frac{dy}{d\theta} + \frac{\partial w}{\partial z}\cdot\frac{dz}{d\theta}$$

$$= \frac{2x}{2\sqrt{x^2+y^2+z^2}}\cdot(-\sin\theta) + \frac{2y}{2\sqrt{x^2+y^2+z^2}}\cdot\cos\theta + \frac{2z}{2\sqrt{x^2+y^2+z^2}}\cdot\sec^2\theta$$

subst. values of x, y, z as $x=\cos\theta$, $y=\sin\theta$, $z=\tan\theta$,

$\frac{dw}{d\theta} = \tan\theta\cdot\sec\theta$. $\therefore$ At $\theta=\pi/4$, $\frac{dw}{d\theta} = \sqrt{2}$

13 (b) Find the local linear $\approx$ L of $f = xyz$ at the point $P(1,2,3)$. Compute the error in approximating f by L at the point $Q(1.001, 2.002, 3.003)$ (5)

Sol: $L(x,y,z) = f(a,b,c) + f_x(a,b,c)(x-a) + f_y(a,b,c)(y-b) + f_z(a,b,c)(z-c)$

$(a,b,c) \to (1,2,3)$; $a=1$, $b=2$, $c=3$

$f_x(1,2,3) = 6$, $f_y(1,2,3) = 3$, $f_z(1,2,3) = 2$

$L(x,y,z) = 6x + 3y + 2z - 12$

$L(1.001, 2.002, 3.003) = 6.018$

$f(1.001, 2.002, 3.003) = 6.018018$

Error $= 0.000018$

14 (a) Locate all relative extrema of $f(x,y) = x^3y^2(12-x-y)$

Sol: $f = 12x^3y^2 - x^4y^2 - x^3y^3$; $f_y = 24x^3y - 2x^4y - 3x^3y^2$

$f_x = 36x^2y^2 - 4x^3y^2 - 3x^2y^3$; $f_{yy} = 24x^3 - 2x^4 - 6x^3y$

$f_{xx} = 72xy^2 - 12x^2y^2 - 6xy^3$; $f_{xy} = 72x^2y - 8x^3y - 9x^2y^2$

$\left.\begin{array}{l} f_x = 0 \\ f_y = 0 \end{array}\right\} \Rightarrow (0,0), (6,4)$ are critical points.

$(0,0)$ is the saddle point, $(6,4)$ is the point of maxima.

14 (b) Let f be a differentiable function of 3 variables and suppose that $w = f(x-y, y-z, z-x)$, S.T $\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$

Sol: Let $r = x-y$, $s = y-z$, $t = z-x$. Then $w \to r,s,t \to x,y,z$

$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial r}\cdot\frac{\partial r}{\partial x} + \frac{\partial w}{\partial s}\cdot\frac{\partial s}{\partial x} + \frac{\partial w}{\partial t}\cdot\frac{\partial t}{\partial x} = \frac{\partial w}{\partial r} + \frac{\partial w}{\partial s}\times 0 + \frac{\partial w}{\partial t}\cdot -1 = \frac{\partial w}{\partial r} - \frac{\partial w}{\partial t}$

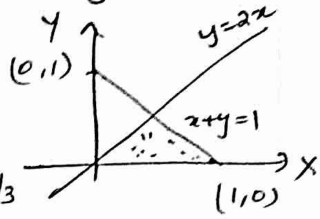
$\frac{\partial w}{\partial y} = \frac{\partial w}{\partial r}\cdot -1 + \frac{\partial w}{\partial s} + \frac{\partial w}{\partial t}\times 0 = -\frac{\partial w}{\partial r} + \frac{\partial w}{\partial s}$

$\frac{\partial w}{\partial z} = \frac{\partial w}{\partial r}\times 0 + \frac{\partial w}{\partial s}\cdot -1 + \frac{\partial w}{\partial t} = -\frac{\partial w}{\partial s} + \frac{\partial w}{\partial t}$

$\frac{\partial w}{\partial x} + \frac{\partial w}{\partial y} + \frac{\partial w}{\partial z} = 0$

15 (a) Find the area bounded by the x-axis, $y = 2x$, $x + y = 1$

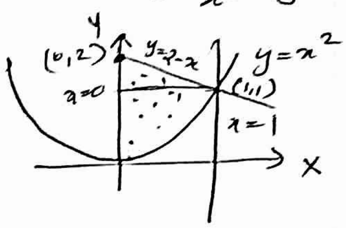
Sol: $A = \iint dx\,dy = \int_{y=0}^{2/3} \int_{x=y/2}^{1-y} dx\,dy$



$= \int_{y=0}^{2/3} \left(1 - y - y/2\right) dy = \left[y - \frac{y^2}{2} - \frac{1}{2}\cdot\frac{y^2}{2}\right]_0^{2/3} = \frac{1}{3}$

15 (b) Change the order of integration and hence solve $\int_0^1 \int_{x^2}^{2-x} dy\,dx$

Sol: $y = x^2$, $y = 2 - x$, $x = 0$, $x = 1$

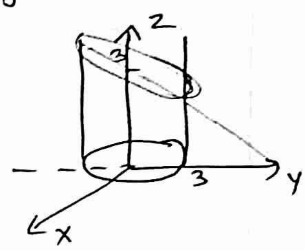


$\int_0^1 \int_{x^2}^{2-x} dy\,dx = \int_{y=0}^{1} \int_{x=0}^{\sqrt{y}} dx\,dy + \int_{y=1}^{2} \int_{x=0}^{2-y} dx\,dy$

$= \int_{y=0}^{1} \sqrt{y}\,dy + \int_{y=1}^{2} (2-y)\,dy = \cdots = \frac{7}{6}$

16 (a) Find the volume bounded by the cylinder $x^2 + y^2 = 9$ and the planes $y + z = 3$ and $z = 0$

$z = 3 - y$



Sol: $V = \iiint dv = \int_{x=-3}^{3} \int_{y=-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_{z=0}^{3-y} dz\,dy\,dx$

$= \int_{x=-3}^{3} \int_{y=-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (3-y)\,dy\,dx = \int_{x=-3}^{3} \left[3y - \frac{y^2}{2}\right]_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} dx$

$= \int_{x=-3}^{3} \left[3\sqrt{9-x^2} - \frac{1}{2}(9-x^2)\right] - \left[3\cdot\sqrt{9-x^2} - \frac{1}{2}(9-x^2)\right] dx$

$= \int_{x=-3}^{3} 6\sqrt{9-x^2}\,dx = 6 \times \frac{\pi}{2} \times 9 = 27\pi$

16 (b) Find the mass and centre of gravity of the lamina in the 1st quadrant bounded by the circle $x^2 + y^2 = 1$ and the cordinate planes with density function xy.

Sol: $M = \iint xy\,dA = \int_0^1 \int_0^{\sqrt{1-x^2}} xy\,dy\,dx = \frac{1}{8}$

$M_y = \iint_R x\cdot(xy)\,dA = \int_0^1 \int_0^{\sqrt{1-x^2}} x^2 y\,dy\,dx = \frac{1}{15}$

$$M_x = \iint_R y(xy)\,dA = \int_0^1 \int_0^{\sqrt{1-x^2}} xy^2\,dy\,dx = \frac{1}{15}$$

$$\bar{x} = \frac{M_y}{M} = \frac{8}{15}, \quad \bar{y} = \frac{M_x}{M} = \frac{8}{15} \qquad \therefore (\bar{x},\bar{y}) = \left(\frac{8}{15}, \frac{8}{15}\right)$$

17 (a) Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{k(k-1)}{(k+1)(k+2)(k+3)}$ (ii) $\sum_{k=1}^{\infty} \left(\frac{k+2}{2k-1}\right)^k$

Sol: (i) $U_k = \frac{k^2(1-1/k)}{k^3(1+1/k)(1+2/k)(1+3/k)} \Rightarrow \sum V_k = \frac{1}{k}$ diverges

$\lim_{k\to\infty} \frac{U_k}{V_k} = 1 \neq 0$ finite $\therefore \sum U_k$ also diverges by limit comparison test.

(ii) $(U_k)^{1/k} = \frac{k+2}{2k-1} \quad \therefore \lim_{k\to\infty} (U_k)^{1/k} = \lim_{k\to\infty} \frac{1+2/k}{k(2-1/k)} = \frac{1}{2} < 1 \Rightarrow$ Converges by root test.

17 (b) Test whether the following series is absolutely convergent or conditionally convergent? $\sum_{k=1}^{\infty} \frac{(-1)^k}{\sqrt{k(k+1)}}$

Sol: $|U_k| = \frac{1}{\sqrt{k(k+1)}} = \frac{1}{k\sqrt{1+1/k}}, \quad \sum V_k = \sum \frac{1}{k}$ diverges.

$\lim_{k\to\infty} \frac{U_k}{V_k} = 1 \neq 0$ finite $\therefore$ By comparison test $\sum |U_k|$ diverges $\Rightarrow$ not abs. convergent

Here; $u_1 > u_2 > u_3 > \cdots$ and $\lim_{k\to\infty} U_k = 0 \quad \therefore$ By Leibnitz's test given series is convergent $\Rightarrow$ Series is conditionally convergent.

18 (a) Test the convergence of the series $1 + \frac{1\cdot 3}{1\cdot 2} + \frac{1\cdot 3\cdot 5}{1\cdot 2\cdot 3} + \frac{1\cdot 3\cdot 5\cdot 7}{1\cdot 2\cdot 3\cdot 4} + \cdots$

Sol: $U_k = \frac{1\cdot 3\cdot 5 \cdots (2k-1)}{k!}, \quad U_{k+1} = \frac{1\cdot 3 \cdots (2k-1)(2k+1)}{(k+1)!}$

$\lim_{k\to\infty} \frac{U_{k+1}}{U_k} = \lim_{k\to\infty} \frac{2k+1}{k+1} = \lim_{k\to\infty} \frac{k(2+1/k)}{k(1+1/k)} = 2 > 0 \Rightarrow$ By ratio test series diverges

18 (b) Test the convergence of (i) $\sum_{k=1}^{\infty} \frac{2}{3^k+5}$ (ii) $\sum_{k=1}^{\infty} (-1)^{k+1} \left(\frac{k}{2k+3}\right)$

Sol: (i) $U_k = \frac{2}{3^k+5}$. Let $\sum V_k = \frac{1}{3^k} = \frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \cdots \quad r = 1/3 \Rightarrow$ Geometric series $\sum \frac{1}{3^k}$ Converges.

$\because \sum U_k < \sum V_k$; $\sum U_k$ also converges. $\boxed{\text{OR}}$ apply ratio test.

(ii) $u_1 = \frac{1}{5}, u_2 = \frac{2}{7}, \cdots \Rightarrow u_1 < u_2 < \cdots \quad \therefore$ By Leibnitz's test alternating series is not convergent.

19(a) Find the fourier series of $f(x) = \begin{cases} 1+\frac{2x}{\pi}, & -\pi < x < 0 \\ 1-\frac{2x}{\pi}, & 0 < x < \pi \end{cases}$

Sol: Given $f(x)$ is an even function. $\therefore b_n = 0$.

$$a_0 = \frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)dx = \cancel{\frac{1}{\pi}\left[\int_{-\pi}^{\pi} 1+\frac{2x}{\pi}\right.} = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x)dx = \frac{2}{\pi}\int_0^{\pi}\left(1-\frac{2x}{\pi}\right)dx$$

$$= \frac{2}{\pi}\left[x - \frac{2x^2}{\pi\, 2}\right]_0^{\pi} = \frac{2}{\pi}\left[\pi - \pi\right] = 0$$

$$a_n = \frac{1}{l}\int_{\alpha}^{\alpha+2l} f(x)\cos\left(\frac{n\pi x}{l}\right)dx = \frac{1}{\pi}\int_{-\pi}^{\pi} f(x).\cos nx\, dx = \frac{2}{\pi}\int_0^{\pi}\left(1-\frac{2x}{\pi}\right)\cos nx\, dx$$

$$= \frac{4(1-(-1)^n)}{n^2\pi^2}$$

$$\therefore f(x) = \sum_{n=1}^{\infty} 4\,\frac{1-(-1)^n}{n^2\pi^2}\cos nx$$

19(b) Obtain fourier series of e^x in $(-1,1)$ $\qquad l = \frac{1-(-1)}{2} = 1$

Sol: $a_0 = \frac{1}{1}\int_{-1}^{1} e^x dx = e^1 - e^{-1}$

$$a_n = 1\int_{-1}^{1} e^x \cos n\pi x\, dx = \frac{(-1)^n}{1+n^2\pi^2}\left(e - \frac{1}{e}\right)$$

$$b_n = 1\int_{-1}^{1} e^x \sin n\pi x\, dx = \frac{n\pi(-1)^n}{1+n^2\pi^2}\left[e^{-1} - e\right]$$

$$\therefore f(x) = \frac{1}{2}(e - e^{-1}) + \sum_{n=1}^{\infty}\left[a_n \cos n\pi x + b_n \sin n\pi x\right]$$ //

20(a) Find the fourier series of $f(x) = x^2 - 2$ in $(-2,2)$

Sol: $f(x)$ is an even fn in $(-2,2)$ $\Rightarrow b_n = 0$

$$a_0 = \frac{1}{2}\int_{-2}^{2} f(x)dx = \frac{1}{2}\times 2\int_0^2 (x^2-2)\,dx = -\frac{4}{3}$$

$$a_n = \frac{1}{2}\int_{-2}^{2} f(x)\cos\frac{n\pi x}{2}dx = \frac{1}{2}\times 2\int_0^2 (x^2-2)\cos\left(\frac{n\pi x}{2}\right)dx = \frac{16(-1)^n}{n^2\pi^2}$$

$$\therefore x^2 - 2 = \frac{1}{2}\left(-\frac{4}{3}\right) + \sum_{n=1}^{\infty}\frac{16(-1)^n}{n^2\pi^2}\cos\left(\frac{n\pi x}{2}\right)$$

20(b) Find the Fourier cosine series of $f(x) = x^2$, $0 < x < \pi$

Sol: $a_0 = \frac{2}{\pi}\int_0^{\pi} x^2\,dx = \frac{2}{3}\pi^2$

$$a_n = \frac{2}{\pi}\int_0^{\pi} x^2\cos nx\,dx = \frac{2}{\pi}\left[x^2\cdot\frac{\sin nx}{n} - 2x\cdot\frac{-\cos nx}{n^2} + 2\cdot\frac{-\sin nx}{n^3}\right]_0^{\pi} = \frac{4(-1)^n}{n^2}$$

$$\therefore f(x) = x^2 = \frac{1}{3}\pi^2 + \sum_{n=1}^{\infty}\frac{4(-1)^n}{n^2}\cos nx$$ //